Roblodex

Discovery & Intelligence Platform

Technical Proposal — V1 Scope of Work Response

Date: April 2026

What We're Building

Roblodex is a data and intelligence system that makes Roblox's 150M+ daily active users discoverable to each other, and in future, makes the resulting signal data invaluable to developers.

The Indexer

Continuous data ingestion — game metadata, time-series metrics, platform sort rankings

The Meaning Machine

LLM-powered taxonomy & semantic search across classification dimensions

The Signal Engine

Accounts, reviews, collections, follows, event tracking

480 hours | 12 weeks | 3 milestones | $28,800 at $60/hr

Milestone-based payments with clear acceptance criteria at each gate

The Discovery Gap

With millions of experience, finding games that match a player's intent is effectively impossible through native tools.

What Players Get Today

Keyword matching. Opaque algorithmic sorts. No way to say "chill anime games I can play solo" and get useful results.

What Roblodex Delivers

Intent-based semantic search, multi-dimensional taxonomy, social signals, and compounding discovery intelligence.

Three Interlocking Layers

- Data Layer - continuously indexes game metadata, time-series metrics, and platform sort rankings
- Intelligence Layer - classifies every game across semantic dimensions, enabling intent-based search
- Social Layer - captures user signals (reviews, collections, follows, events) feeding a compounding data flywheel

Rotrends: Strengths & Weaknesses

Where They Lead

228K+ games	Down to 0 CCU
10-min refresh	For popular games
34+ metrics	With historical data
Change detection	Metadata versioning

Where They're Weak

- Search is keyword-only - no semantic understanding
- No social features - read-only analytics
- Game pages are pure stats - no context
- Collections gated behind $15/mo Pro
- Spreadsheet-first design

"This is how it should have been."

— Target reaction from a Rotrends user visiting Roblodex

Technology Stack

Frontend	**Next.js (App Router, TS)**	SSR for SEO, ISR for game pages, clean slug routing
Backend	**Node.js (TypeScript)**	Shared language, modular services, strong async
Database	**PostgreSQL (managed)**	Flexible, JSON support, TimescaleDB compatible
Time-Series	**TimescaleDB**	Same Postgres instance, rollups, 10-min granularity
Search	**Typesense**	Tag intersection, synonyms, fuzzy match, autocomplete
Jobs	**BullMQ (Redis)**	Scheduled polling, retries, rate limiting, priorities
Visualisation	**D3.js + Recharts**	D3 for bespoke visuals, Recharts for standard charts

Infrastructure

Component	Provider	Rationale
Frontend hosting	**Vercel**	Native Next.js, global CDN, zero-config
Backend + workers	**AWS (ECS / EC2)**	Scalable compute for 24/7 polling
Database	**Supabase / RDS**	Managed Postgres + TimescaleDB
Search index	**Typesense Cloud**	Managed, auto-scaling
Redis	**Upstash / ElastiCache**	Serverless for BullMQ queues
LLM	**OpenAI API**	GPT-4o-mini for classification

Indexer Architecture

Data Ingestion

- Game metadata (title, desc, creator, genre, thumbnails, passes)
- Time-series: CCU, visits, favourites, ratings
- Platform sort rankings with full history
- Metadata versioning & change detection

On-Demand Indexing

When a user searches for a valid Roblox game not in the catalogue, the system triggers immediate indexing. Game is pulled, classified, and searchable within seconds - growing the catalogue organically.

Tiered Refresh Architecture

Tier	Games	Refresh	Trigger
Hot	Top 500 by CCU	Every 10 min	CCU threshold
Warm	Mid 2K–5K	30 min – daily	Popularity band
Cold	Long-tail	Weekly / on-demand	When searched

Search Architecture

LLM Classification: OpenAI GPT-4o-mini | ~$0.001–0.003 per game | 10K games ≈ $10–30

Query Decomposition	"anime fighting games" → intent {anime-like, fighting-like} → match both
Tag Intersection	Multiple attribute filters applied simultaneously — core search mechanic
Synonym Expansion	"brawler" also surfaces fighting, PvP, combat results
Fuzzy Matching	"simualtor" still finds simulators (Typesense native)
Autocomplete	Across game names, tags, and collections — sub-200ms response
Explainability	Results surface why they matched — which tags, how strongly
Faceted Filtering	Results filterable by any taxonomy dimension

Signal Engine Features

Accounts & Onboarding

Roblox OAuth only. Three layers:
- L1 (auto): Sign in → import history → taste profile
- L2 (guided): Archetype selection
- L3 (fun): Play style self-perception

Structured Reviews

Tap-to-select, not free-text:
- "What's the vibe?" → select tags
- "Who should play this?" → player types
- Verified Player badge from play history
- Every review feeds taxonomy

Collections & Follow

Free & unlimited (vs Rotrends $15/mo Pro).
Public by default, shareable URLs.
Follow curators, games, or tags.
In-app notifications V1, email/push V1.5.

Taste Cards (Viral)

Auto-generated visual cards from user activity.
Designed for Discord/Twitter/TikTok sharing.
Spotify Wrapped model — free marketing.
Links back to Roblodex profile.

Event Tracking — Day-One Infrastructure

Every user action captured as a structured event from launch — this is the fuel for V1.5's developer analytics tier.

- Searches (query, filters, clicks)
- Saves & likes
- Collection additions
- Review submissions
- Click-through to Roblox
- Time on game page
- Follow actions

API Confidence & Validation Plan

Endpoint	Confidence
Game metadata	High
Player counts / CCU	High
Thumbnails API	High
Sort rankings	Medium
User data (OAuth)	High
Play history (OAuth)	Low — must validate
Favourites (OAuth)	Medium

Week 1 Validation Plan

- Controlled load tests per endpoint category
- Establish safe request rates with margin
- Design polling schedule around confirmed limits
- Register OAuth app, document available scopes
- Assess IP rotation needs for scale
- Deliver findings report before M1 delivery

Milestone 1 — Core Data Layer

Weeks 1-3 | 120 hours | $7,200

Deliverables

- Database schema (Postgres + TimescaleDB)
- Roblox API integration — 5K–10K games
- Tiered polling system (hot/warm/cold) 24/7
- On-demand indexing endpoint
- Metadata versioning & change detection
- Sort ranking tracking with history
- API rate limit & OAuth scope reports

Acceptance Criteria

- 5K+ games indexed with full metadata
- Time-series collecting at correct intervals
- On-demand index < 10 seconds
- Change detection demonstrated
- API findings documented
- OAuth scope report with implications

Milestone 2 — Meaning Machine + Search

Weeks 4-8 | 200 hours | $12,000

Deliverables

- LLM classification pipeline (batch)
- Full taxonomy — all 8 dimensions
- Synonym graphs (seeded + expandable)
- Typesense search index live
- Query decomposition logic
- Autocomplete (games, tags, collections)
- Search explainability
- Draft wireframes for key pages

Acceptance Criteria

- All games classified across all dimensions
- "chill anime games solo" returns relevant results
- Tag intersection works correctly
- Typo tolerance: "simualtor" → simulators
- Autocomplete < 200ms
- Wireframes reviewed & approved

Milestone 3 — Product Layer

Weeks 9-12 | 160 hours | $9,600

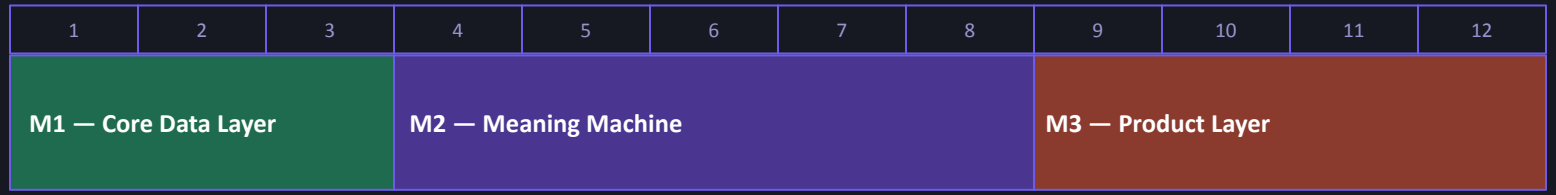
Deliverables

- OAuth sign-in + 3-layer onboarding
- Structured reviews (tap-to-select, badges)
- Collections (free, unlimited, shareable)
- Taste Cards (auto-generated, viral)
- Follow mechanics + in-app notifications
- Full game pages with all content
- All additional pages (home, discover, etc.)
- Event tracking pipeline
- SEO: SSR, structured data, OG tags
- Responsive + accessibility baseline

Acceptance Criteria

- Full user journey testable end-to-end
- SSR + structured data verified
- Taste Card renders on Discord/Twitter
- All event types captured
- Mobile responsive tested
- Production deployment live & stable

Timeline Overview



Key Dates

Week 1	API rate limit validation + OAuth scope report
Week 3	Milestone 1 delivery — data layer complete
Week 6	Wireframes delivered for review
Week 8	Milestone 2 delivery — search system live
Week 10	Internal testing / polish sprint
Week 12	Milestone 3 delivery — full product launch

Budget Breakdown

$28,800	480h	$60/hr	12 wks
Total Budget	Total Hours	Hourly Rate	Timeline

Milestone	Hours	Cost	Payment Trigger
M1 — Core Data Layer	120h	**$7,200**	Data layer live, API report delivered
M2 — Meaning Machine	200h	**$12,000**	Search live, wireframes approved
M3 — Product Layer	160h	**$9,600**	Full launch, all criteria met
Total	**480h**	**$28,800**	

Infrastructure Costs (Post-Launch)

Launch

5K Games

$150-300/mo

Vercel + small EC2 + Supabase free/pro

Growth

25K Games

$400-900/mo

Larger compute, Typesense Cloud, more storage

Scale

100K+ Games

$1.2K-3.5K/mo

Multi-worker, larger DB, higher LLM volume

Breakdown by Service

Service	Launch	Growth	Scale
Vercel	$20	$20	$20–50
AWS compute	$50–100	$150–400	$500–1.5K
Database	$25–50	$100–250	$300–800
Typesense	$30–60	$60–120	$200–500
Redis	$10–20	$20–50	$50–150
LLM	$5–15	$15–40	$50–200

Risks & Mitigations

API rate limits tighter than expected Likelihood: Med	Week 1 validation. Fallback: reduce frequency, batch, add IP rotation.
OAuth doesn't expose play history Likelihood: Med-High	Week 1 assessment. Fallback: simplified onboarding, on-platform activity.
Search quality needs iteration Likelihood: High	Expected. M2 delivers baseline; synonyms + reviews improve over time.
Scope density — 12 weeks is tight Likelihood: Med	Milestone gates. Trade-offs proposed, not silent quality cuts.
Roblox API changes/deprecation Likelihood: Low	Adapter architecture isolates API code. One module to update.
LLM classification insufficient Likelihood: Low-Med	GPT-4o-mini strong. Fallback: upgrade to GPT-4o. Reviews correct errors.